On Tensorized Φ-Strong Data Processing Inequality Constants

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Abstract

In this paper, we prove that for a bivariate source $(X, Y) \sim P_{XY}$ with X taking binary values, the standard strong data processing inequality (SDPI) constant (i.e., the hypercontractivity constant) is the maximum among all Φ-SDPI constants for twice continuously differentiable, convex Φ such that $\frac{1}{\Phi''}$ is concave. This class of functions Φ's was widely considered in the literature to ensure the tensorization of Φ-SDPI constants. As a consequence, all Φ-SDPI constants of an arbitrary channel for this class of Φ's are identical.

Index Terms

Strong data processing inequality, hypercontractivity inequality, tensorization, subadditivity.

I. Introduction

Let $(X, Y) \sim P_{XY}$ be a pair of random variables defined on $\mathcal{X} \times \mathcal{Y}$. Let $f : \mathcal{X} \to \mathbb{R}_{\geq 0}$, and $f(X)$ a random variable induced by X. Let $\Phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$. Then the Φ-entropy of f is defined by

$$H_\Phi(f) = \mathbb{E}[\Phi(f)] - \Phi(\mathbb{E}f).$$

In this paper we always assume that Φ is convex, in which case

$$H_\Phi(f) \geq 0,$$

by Jensen's inequality. For the choice of $\Phi(t) = t^2 - 1$, the Φ-entropy reduces to variance: $H_\Phi(f) = \mathrm{Var}(f)$; for the choice of $\Phi(t) = t \log t$, the Φ-entropy reduces to the relative entropy (when $\mathbb{E}[f] = 1$).

Definition 1. Let $\mathcal{F}$ be the class of all twice continuously differentiable functions $\Phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$ such that $\Phi'' > 0$ (i.e., Φ is convex), $\frac{1}{\Phi''}$ is concave, and $\Phi(1) = 0$ (which is assumed WLOG). Let $\widetilde{\mathcal{F}}$ be the class of all non-constant functions $f : \mathcal{X} \to \mathbb{R}_{\geq 0}$ such that $\mathbb{E}[f] = 1$. In this case, $H_\Phi(f) = \mathbb{E}[\Phi(f)]$ for $f \in \widetilde{\mathcal{F}}$.

Definition 2. The Φ-strong data processing inequality constant (Φ-SDPI constant or Φ-entropy contraction coefficient) for a source pair $(X, Y) \sim P_{XY}$ is defined as

$$\eta_\Phi(X, Y) := \eta_\Phi(P_{XY}) := \sup_{f \in \widetilde{\mathcal{F}}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}.$$

Any $f \in \widetilde{\mathcal{F}}$ can be seen as the density of a distribution Q_X with respect to the distribution P_X, and in this case, $y \mapsto \mathbb{E}[f|Y = y]$ will be the density of a distribution Q_Y with respect to the distribution P_Y, where Q_Y is the output distribution of the channel $P_{Y|X}$ when the input is Q_X. Denote

$$D_\Phi(Q\|P) = \int \Phi(\frac{dQ}{dP}) dQ,$$

which is known as the "f-divergence". Then, we can rewrite

$$H_\Phi(f) = D_\Phi(Q_X\|P_X),$$
$$H_\Phi(\mathbb{E}[f|Y]) = D_\Phi(Q_Y\|P_Y),$$

and

$$\eta_\Phi(X, Y) = \sup_{Q_X : Q_X \neq P_X} \frac{D_\Phi(Q_Y\|P_Y)}{D_\Phi(Q_X\|P_X)}. \tag{1}$$

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Thus, the Φ-SDPI constant $\eta_\Phi(X,Y)$ characterizes the maximum possible ratio between the "f-divergence" of two channel inputs and the "f-divergence" of the corresponding channel outputs.

The Φ-SDPI constant was recently systematically studied in [1], [2] for discrete distributions and in [3]–[5] for continuous distributions or distributions defined on abstract alphabets. A special case of the Φ-SDPI constant—known as the SDPI constant defined by

$$\eta_{\mathrm{KL}}(X,Y) = \sup_{Q_X : Q_X \neq P_X} \frac{D(Q_Y \| P_Y)}{D(Q_X \| P_X)}, \tag{2}$$

was first introduced by Ahlswede and Gács in a landmark paper [6]. Here D is the relative entropy. For this case, the SDPI constant is known to be equal to the hypercontractivity constant, with the latter defined as follows.

Definition 3. For $p \geq 1$ define

$$q_p^*(X,Y) := \inf \left\{ q : \|\mathbb{E}[f(X)|Y]\|_p \leq \|f(X)\|_q, \forall f : \mathcal{X} \to \mathbb{R}_{\geq 0} \right\}.$$

Moreover, define the hypercontractivity constant $s^*(X,Y)$ as

$$s^*(X,Y) := \lim_{p\to\infty} \frac{q_p^*(X,Y) - 1}{p-1}.$$

It is known that $\eta_{\mathrm{KL}}(X,Y) = s^*(X,Y)$ [6].

For Φ-entropy, it is usually assumed that Φ is convex. However, when the tensorization property is required, then it is common to restrict $\Phi \in \mathcal{F}$ [1]. In fact, assume Φ is twice differentiable with $\Phi'' > 0$, then the condition $\Phi \in \mathcal{F}$ is necessary and sufficient for Φ-entropy to be subadditive [7, Theorem 14.1 and Proposition 14.3]. Subadditivity of Φ-entropy implies the tensorization property of Φ-SDPI constant [1]. Throughout this paper, we focus on the case $\Phi \in \mathcal{F}$, i.e., we focus on the Φ-SDPI which satisfies the tensorization property.

From the convexity of Φ, $\eta_\Phi(X,Y) \in [0,1]$. Moreover, if X and Y are independent, then $\eta_\Phi(X,Y) = 0$. If X and Y have common randomness (i.e., $f(X) = g(Y)$ for some non-constant f and g), then $\eta_\Phi(X,Y) = 1$. The doubly symmetric binary source (DSBS) with correlation ρ, denoted by $\mathrm{DSBS}(\rho)$, is defined as a pair of binary random variables (X,Y) such that

$$\mathbb{P}(X=0,Y=0) = \mathbb{P}(X=1,Y=1) = \frac{1+\rho}{4},$$
$$\mathbb{P}(X=0,Y=1) = \mathbb{P}(X=1,Y=0) = \frac{1-\rho}{4}.$$

It is known that [1], [2] for the DSBS, $\eta_\Phi(X,Y) = \rho^2$ holds for all $\Phi \in \mathcal{F}$.

When $\Phi(t) = t^2 - 1$, $\eta_\Phi(X,Y)$ is exactly $\rho_{\mathrm{m}}^2(X,Y)$ as shown in [8], where $\rho_{\mathrm{m}}(X,Y)$ denotes the maximal correlation between X, Y. Moreover, for the choice of $\Phi(t) = t\log t$, $\eta_\Phi(X,Y) = \eta_{\mathrm{KL}}(X,Y)$ is known to be equal to the hypercontractivity constant $s^*(X,Y)$ [6]. Hence the maximal correlation and the hypercontractivity constant can be seen as special cases of Φ-SDPI constants. For all these constants, they all satisfy the following data processing inequality: for any $U \to X \to Y \to V$,

$$\eta_\Phi(U,V) \leq \eta_\Phi(X,Y) \tag{3}$$

for any convex Φ. The hypercontractivity inequality has been widely applied in various fields, including discrete Fourier analysis, combinatorics, probability theory, and information theory; please refer to [6], [9]–[12]. A natural question arises: Can we apply the Φ-SDPI to strengthen the hypercontractivity inequality by finding a better Φ? In this paper, we show that when (X,Y) is the DSBS and (U,V) is a pair of binary random variables, the hypercontractivity inequality is the strongest Φ-SDPI.

II. Main Results

We next state our results. It is well-known that $\rho_{\mathrm{m}}^2(X,Y) \leq s^*(X,Y)$, where $\rho_{\mathrm{m}}(X,Y)$ is the maximal correlation. Moreover, Raginsky [1] proved that $\rho_{\mathrm{m}}^2(X,Y) \leq \eta_\Phi(X,Y)$ for any $\Phi \in \mathcal{F}$. The following theorem shows that $\eta_\Phi(X,Y) \leq s^*(X,Y)$ for binary X.

Theorem 1. *If X is binary, then*

$$\sup_{\Phi\in\mathcal{F}} \eta_\Phi(X,Y) = \eta_{\mathrm{KL}}(X,Y) = s^*(X,Y).$$

Proof: We will prove this theorem for the case of finite Y in Section IV. We can easily extend the result to the case that Y has abstract alphabet (but X is still binary) as follows. Let $\mathcal{P}$ be a finite measurable partition of $\mathcal{Y}$, and denote $Y_{\mathcal{P}}$

as the output random variable of the partition $\mathcal{P}$ (considered as a channel) when the input is Y. It is already known that $D_\Phi(Q_Y\|P_Y) = \sup_{\mathcal{P}} D_\Phi(Q_{Y_{\mathcal{P}}}\|P_{Y_{\mathcal{P}}})$. We then have that

$$\begin{aligned}
\sup_{\Phi\in\mathcal{F}} \eta_\Phi(X,Y) &= \sup_{\Phi\in\mathcal{F}} \sup_{Q_X:Q_X\neq P_X} \frac{D_\Phi(Q_Y\|P_Y)}{D_\Phi(Q_X\|P_X)} \\
&= \sup_{\Phi\in\mathcal{F}} \sup_{Q_X:Q_X\neq P_X} \frac{\sup_{\mathcal{P}} D_\Phi(Q_{Y_{\mathcal{P}}}\|P_{Y_{\mathcal{P}}})}{D_\Phi(Q_X\|P_X)} \\
&= \sup_{\mathcal{P}} \sup_{\Phi\in\mathcal{F}} \sup_{Q_X:Q_X\neq P_X} \frac{D_\Phi(Q_{Y_{\mathcal{P}}}\|P_{Y_{\mathcal{P}}})}{D_\Phi(Q_X\|P_X)} \\
&= \sup_{\mathcal{P}} \sup_{Q_X:Q_X\neq P_X} \frac{D(Q_{Y_{\mathcal{P}}}\|P_{Y_{\mathcal{P}}})}{D(Q_X\|P_X)} \qquad (4)\\
&= \sup_{\Phi\in\mathcal{F}} \eta_{\mathrm{KL}}(X,Y),
\end{aligned}$$

where (4) follows by our theorem for the case of finite Y. ■

Combining this result with the fact that $\eta_\Phi(X,Y) \geq \rho_{\mathrm{m}}^2(X,Y)$ [1, Theorem III.3] [4, Theorem 2] [2] yields that for binary X and any $\Phi \in \mathcal{F}$,

$$\rho_{\mathrm{m}}^2(X,Y) \leq \eta_\Phi(X,Y) \leq \eta_{\mathrm{KL}}(X,Y). \qquad (5)$$

Both the bounds are sharp. The lower bound is attained by $\Phi(t) = t^2 - 1$; the upper bound is attained by $\Phi(t) = t\log t$. Furthermore, even if we relax Φ to be twice differentiable and convex such that $\Phi''(1) > 0$ (but $\frac{1}{\Phi''}$ is not necessarily to be concave), then the sharp lower bound on $\eta_\Phi(X,Y)$ is still $\rho_{\mathrm{m}}^2(X,Y)$. However, if we require Φ to be operationally convex (not necessarily belong to $\mathcal{F}$), the sharp upper bound is $\max\left\{\rho_{\mathrm{m}}^2(X,Y), \sup_{0<\lambda<1} \eta_{\mathrm{LC}_\lambda}(X,Y)\right\}$ where the LC_λ-divergence with $\mathrm{LC}_\lambda(t) := 1 - \frac{t}{\lambda t+1-\lambda}$ corresponds to the Le Cam divergence with parameter λ [1, Theorem III.6].

If $(X,Y) \sim \mathrm{DSBS}(\rho)$, then (3) turns into

$$\eta_\Phi(U,V) \leq \rho^2 \qquad (6)$$

for all $\Phi \in \mathcal{F}$. Since $\sup_{\Phi\in\mathcal{F}} \eta_\Phi(U,V) \leq s^*(U,V)$, the strongest form of (6) over all choices $\Phi \in \mathcal{F}$ is $s^*(U,V) \leq \rho^2$ given by hypercontractivity inequalities. Hence hypercontractivity inequalities are the best strong data processing inequalities for the DSBS in the sense that any other strong data processing inequalities are implied by hypercontractivity inequalities.

It is natural to extend Theorem 1 to the general case with X defined on an arbitrary alphabet. We pose the following question.

Problem 1. Is it true that $\sup_{\Phi\in\mathcal{F}} \eta_\Phi(X,Y) = s^*(X,Y)$ for any X, Y defined on arbitrary alphabets?

The answer is yes, if one can show that for any X, Y defined on arbitrary alphabets, given $f \in \widetilde{\mathcal{F}}$,

$$\sup_{g\in\mathcal{G}} \frac{H_g(\mathbb{E}[f|Y])}{H_g(f)}$$

is attained by $g(t) = t$ or $g(t) = 1$.

Recall that in the definition of $\eta_\Phi(X,Y)$, the input distribution P_X is fixed. However, in the literature, it is also common to consider the Φ-SDPI constant for the *channel* $P_{Y|X}$ in which we optimize over P_X as well.

Definition 4. The Φ-strong data processing inequality constant (Φ-SDPI constant or Φ-entropy contraction coefficient) for the channel $P_{Y|X}$ is defined as

$$\eta_\Phi(P_{Y|X}) := \sup_{P_X} \eta_\Phi(P_{XY}) = \sup_{P_X,Q_X:Q_X\neq P_X} \frac{D_\Phi(Q_Y\|P_Y)}{D_\Phi(Q_X\|P_X)}. \qquad (7)$$

It was shown by Ordentlich and Polyanskiy [13] that the optimization above can be restricted to pairs (P_X, Q_X) supported on two common points in $\mathcal{X}$ (i.e., P_X, Q_X can be assumed to be two Bernoulli distributions). Combining this with our main result in this paper, we obtain the following interesting result, which states that all Φ-SDPI constants for a given channel are identical for all $\Phi \in \mathcal{F}$ (roughly speaking, for all Φ's making the Φ-SDPI tensorized).

Theorem 2. *For any channel $P_{Y|X}$, the Φ-SDPI constants $\eta_\Phi(P_{Y|X})$ for all $\Phi \in \mathcal{F}$ are identical, i.e.,*

$$\eta_\Phi(P_{Y|X}) = \eta_{\mathrm{KL}}(P_{Y|X}) = \eta_{\chi^2}(P_{Y|X}),\ \forall \Phi \in \mathcal{F},$$

where

$$\eta_{\mathrm{KL}}(P_{Y|X}) := \sup_{P_X} \eta_{\mathrm{KL}}(P_{XY}) = \sup_{P_X,Q_X:Q_X\neq P_X} \frac{D(Q_Y\|P_Y)}{D(Q_X\|P_X)}$$

is the standard SDPI constant (or contraction coefficient) for the channel $P_{Y|X}$, and $\eta_{\chi^2}(P_{Y|X})$ is the $\eta_\Phi(P_{Y|X})$ with $\Phi(t) = t^2 - 1$ and is also equal to $\sup_{P_X} \rho_{\mathrm{m}}(X,Y)$.

Proof: By [13, Theorem 2], without loss of optimality, we can assume that $\mathcal{X}$ is a binary set in $\eta_\Phi(P_{Y|X}) = \sup_{P_X} \eta_\Phi(P_{XY})$. By Theorem 1, in this case, for $\Phi \in \mathcal{F}$, $\eta_\Phi(P_{XY}) \leq \eta_{\mathrm{KL}}(P_{XY})$. Therefore,

$$\eta_\Phi(P_{Y|X}) \leq \eta_{\mathrm{KL}}(P_{Y|X}),\ \forall \Phi \in \mathcal{F}.$$

On the other hand, by perturbation analysis, $\eta_\Phi(P_{Y|X}) \geq \eta_{\chi^2}(P_{Y|X})$ [1, Theorem III.3] [4, Theorem 2]. Moreover, it is known that $\eta_{\chi^2}(P_{Y|X}) = \eta_{\mathrm{KL}}(P_{Y|X})$ always holds [4], [6]. Thus, Theorem 2 follows. ■

We now provide an application of Theorem 2 to analyzing convergence of a Markov process. The Φ-SDPI constants is naturally used to analyze the convergence of a Markov process. Consider $P_{X_1|X_0}$ as the transition probability kernel from X_t to X_{t+1} for all $t \in \mathbb{N}$. Let P_X be the stationary distribution. Then,

$$\begin{aligned}\frac{D_\Phi(P_{X_t}\|P_X)}{D_\Phi(P_{X_0}\|P_X)} &= \frac{D_\Phi(P_{X_1}\|P_X)}{D_\Phi(P_{X_0}\|P_X)}\frac{D_\Phi(P_{X_2}\|P_X)}{D_\Phi(P_{X_1}\|P_X)}\cdots\frac{D_\Phi(P_{X_t}\|P_X)}{D_\Phi(P_{X_{t-1}}\|P_X)}\\ &\leq \eta_\Phi^t(P_{X_1|X_0}),\end{aligned}$$

which characterizes how fast the process converges to P_X in the Φ-divergence. When we consider the vector case and focus on a product kernel $P_{X_1|X_0}^{\otimes n}$ with stationary distribution $P_X^{\otimes n}$. The convergence of the process $\mathbf{X}_t = (X_{1,t}, X_{2,t}, ..., X_{n,t})$ induced by the kernel $P_{X_1|X_0}^{\otimes n}$ to $P_X^{\otimes n}$ in the Φ-divergence is characterized by

$$\frac{D_\Phi(P_{\mathbf{X}_t}\|P_X^{\otimes n})}{D_\Phi(P_{\mathbf{X}_0}\|P_X^{\otimes n})} \leq \eta_\Phi^t(P_{X_1|X_0}^{\otimes n}).$$

When we focus on $\Phi \in \mathcal{F}$. Then, $\eta_\Phi(P_{X_1|X_0}^{\otimes n}) = \eta_\Phi(P_{X_1|X_0})$. The upper bound reduces to $\eta_\Phi^t(P_{X_1|X_0})$ which is dimension-independent. Theorem 2 implies that in this case, the upper bound remains identical for $\Phi \in \mathcal{F}$. Roughly speaking, all Φ-divergence for $\Phi \in \mathcal{F}$ vanishes in the same rate.

III. Preliminaries

WLOG, one can restrict Φ to be a function such that

$$\Phi(1) = 0 \tag{8}$$
$$\Phi'(1) = 0 \tag{9}$$
$$\Phi''(1) = 1, \tag{10}$$

since for fixed f, $\frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ does not change if we replace $\Phi(t)$ with $\alpha\Phi(t) + \beta t + \gamma$ for $\alpha > 0, \beta, \gamma \in \mathbb{R}$. Under the conditions (8)-(10), $\Phi \in \mathcal{F}$ is uniquely determined by $\frac{1}{\Phi''}$. Let $\mathcal{G}$ be the class of all concave functions $g : \mathbb{R}_{\geq 0} \to \mathbb{R}_{>0}$. Equivalent definitions of $\mathcal{G}$ can be found in [7, Exercise 14.2]. By definition, we know that any $g \in \mathcal{G}$ is nondecreasing, and moreover,

$$\mathcal{F} = \left\{ r \in \mathbb{R}_{\geq 0} \mapsto \int_1^r \int_1^s \frac{1}{g(t)} dt ds \in \mathbb{R}_{\geq 0} : g \in \mathcal{G} \right\}.$$

Without ambiguity, we also denote $H_\Phi(f)$ as $H_g(f)$. We next provide several properties of sets $\mathcal{F}$ and $\mathcal{G}$, the function Φ, and the Φ-SDPI constant.

Lemma 1. *Both $\mathcal{F}$ and $\mathcal{G}$ are convex.*

Proof: For $\Phi_1, \Phi_2 \in \mathcal{F}$ and $0 \leq \lambda \leq 1$, denote $\Psi_\lambda = \lambda\Phi_1 + (1-\lambda)\Phi_2$. Then it is easy to verify $\Psi_\lambda'' > 0$. Hence it remains to prove that $\frac{1}{\Psi_\lambda''}$ is concave. For a twice differentiable function Φ, $\frac{1}{\Phi''}$ is concave if and only if $(s,t) \mapsto p\Phi(s) + (1-p)\Phi(t) - \Phi(ps + (1-p)t)$, for any $p \in [0,1]$, is jointly convex. Hence it follows that $\frac{1}{\Psi_\lambda''}$ is concave. ■

Lemma 2. *Given f, $\Phi \in \mathcal{F} \mapsto \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} \in \mathbb{R}_{>0}$ is a quasiconvex function.*

Proof: By definition, H_Φ is a linear functional in Φ. That is, for $\Psi = \lambda\Phi_1 + (1-\lambda)\Phi_2$,

$$H_\Psi(f) = \lambda H_{\Phi_1}(f) + (1-\lambda) H_{\Phi_2}(f).$$

Hence

$$\begin{aligned}\frac{H_\Psi(\mathbb{E}[f|Y])}{H_\Psi(f)} &= \frac{\lambda H_{\Phi_1}(\mathbb{E}[f|Y]) + (1-\lambda) H_{\Phi_2}(\mathbb{E}[f|Y])}{\lambda H_{\Phi_1}(f) + (1-\lambda) H_{\Phi_2}(f)}\\ &\leq \max\left\{\frac{H_{\Phi_1}(\mathbb{E}[f|Y])}{H_{\Phi_1}(f)}, \frac{H_{\Phi_2}(\mathbb{E}[f|Y])}{H_{\Phi_2}(f)}\right\}\end{aligned}$$

That is, given f, $\Phi \in \mathcal{F} \mapsto \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} \in \mathbb{R}_{>0}$ is a quasiconvex function. ■

Lemma 3. *Given f, $\sup_{\Phi\in\mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ is attained by some Φ.*

Proof: WLOG, one can restrict Φ to be a function such that

$$\Phi(1) = 0 \tag{11}$$
$$\Phi'(1) = 0 \tag{12}$$
$$\Phi''(1) = 1, \tag{13}$$

since for fixed f, $\frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ does not change if we replace $\Phi(t)$ with $\alpha\Phi(t) + \beta t + \gamma$ for $\alpha > 0, \beta, \gamma \in \mathbb{R}$. Obviously, in this setting, Φ is nonnegative, and uniquely determined by Φ''. Since 1) Φ'' is positive on $\mathbb{R}_{>0}$, 2) $\frac{1}{\Phi''(1)} = 1$, and 3) $\frac{1}{\Phi''}$ is concave on $\mathbb{R}_{>0}$, we have

$$1 \le \frac{1}{\Phi''(t)} \le t, \forall t \in [1, \infty)$$
$$t \le \frac{1}{\Phi''(t)} \le 1, \forall t \in (0, 1)$$

i.e.,

$$\frac{1}{t} \le \Phi''(t) \le 1, \forall t \in [1, \infty)$$
$$1 \le \Phi''(t) \le \frac{1}{t}, \forall t \in (0, 1)$$

Denote $A = \{f(x_1), ..., f(x_M), \mathbb{E}[f|y_1], .., \mathbb{E}[f|y_N]\}$. By definition, $0 < a_{\min} := \min A < a_{\max} := \max A < +\infty$. Observe that $\frac{1}{a_{\max}} \le \Phi''(t) \le \frac{1}{a_{\min}}$ for all $t \in [a_{\min}, a_{\max}]$.

Obviously, $g := \frac{1}{\Phi''}$ is positive, concave, continuous, increasing on $\mathbb{R}_{>0}$. Hence

$$R_g(x_1, x_2) := \frac{g(x_1) - g(x_2)}{x_1 - x_2}$$

is nonnegative, and monotonically non-increasing in x_1 for every fixed x_2, (or vice versa). Hence for $x_1, x_2 \in [a_{\min}, a_{\max}]$,

$$R_g(x_1, x_2) \le \frac{g(a_{\min}) - g(x_2)}{a_{\min} - x_2} \le \frac{g(a_{\min}) - g(0)}{a_{\min} - 0} \le \frac{g(a_{\min})}{a_{\min}} \le \frac{1}{a_{\min}}.$$

Hence $R_g(x_1, x_2)$ is uniformly bounded on $[a_{\min}, a_{\max}]$ for all functions $g = \frac{1}{\Phi''}$ with $\Phi \in \mathcal{F}$. Observe that

$$R_{\Phi''}(x_1, x_2) = \frac{\Phi''(x_1) - \Phi''(x_2)}{x_1 - x_2} = \frac{\frac{1}{g(x_1)} - \frac{1}{g(x_2)}}{x_1 - x_2} = -\frac{R_g(x_1, x_2)}{g(x_1)g(x_2)}.$$

Hence

$$-\frac{1}{a_{\min}^3} \le R_{\Phi''}(x_1, x_2) \le 0.$$

which implies that $R_{\Phi''}(x_1, x_2)$ is also uniformly bounded on $[a_{\min}, a_{\max}]$ for all functions $\Phi \in \mathcal{F}$.

Given f, let $\{\Phi_n\}$ be a sequence of functions that attains $\sup_{\Phi\in\mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$. Since $\Phi_n \in \mathcal{F}$, $R_{g_n}(x_1, x_2)$ with $g_n := \frac{1}{\Phi_n''}$ is uniformly bounded on $[a_{\min}, a_{\max}]$. This means that the sequence $\{g_n\}$ is uniformly equicontinuous. By the Arzelà–Ascoli theorem, there exists a subsequence $\{g_{n_k}\}_{k\in\mathbb{N}}$ that converges uniformly on $[a_{\min}, a_{\max}]$. Let $\widehat{g}$ be the limit of this subsequence. Then $\frac{1}{g_{n_k}} \to \frac{1}{\widehat{g}}$ and $\Phi_{n_k} \to \widehat{\Phi}$ uniformly on $[a_{\min}, a_{\max}]$, where $\widehat{\Phi}$ is the twice-integral of $\frac{1}{\widehat{g}}$ satisfying (11)-(13). This means

$$\sup_{\Phi\in\mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} = \lim_{k\to\infty} \frac{H_{\Phi_{n_k}}(\mathbb{E}[f|Y])}{H_{\Phi_{n_k}}(f)} = \frac{H_{\widehat{\Phi}}(\mathbb{E}[f|Y])}{H_{\widehat{\Phi}}(f)}.$$

We next construct an extension $\widetilde{\Phi}$ of $\widehat{\Phi}$ to the domain $\mathbb{R}_{>0}$ such that $\widetilde{\Phi} \in \mathcal{F}$. Obviously, $\widehat{g}$ is concave, and it satisfies $a_{\min} \le \widehat{g}(t) \le a_{\max}$ for all $t \in [a_{\min}, a_{\max}]$. Furthermore, for all $x_1, x_2 \in [a_{\min}, a_{\max}]$,

$$0 \le R_{\widehat{g}}(x_1, x_2) \le \frac{\widehat{g}(a_{\min})}{a_{\min}}. \tag{14}$$

We now construct an extension of $\widehat{g}$ as

$$\widetilde{g}(t) = \begin{cases} \widehat{g}(t) & t \in [a_{\min}, a_{\max}] \\ \frac{\widehat{g}(a_{\min})}{a_{\min}} t & t \in (0, a_{\min}) \\ \widehat{g}(a_{\max}) & t \in (a_{\max}, +\infty) \end{cases}$$

Obviously, $\widetilde{g} > 0$. Moreover, by the definition of $\widetilde{g}$, inequalities in (14), and the fact that $\widehat{g}$ is concave, we have that $\widetilde{g}$ is concave. Define $\widetilde{\Phi}$ as the twice-integral of $\frac{1}{\widetilde{g}}$ satisfying (11)-(13). Then obviously, $\widetilde{\Phi} \in \mathcal{F}$ and

$$\frac{H_{\widetilde{\Phi}}(\mathbb{E}[f|Y])}{H_{\widetilde{\Phi}}(f)} = \frac{H_{\widehat{\Phi}}(\mathbb{E}[f|Y])}{H_{\widehat{\Phi}}(f)} = \sup_{\Phi \in \mathcal{F}} \frac{H_{\Phi}(\mathbb{E}[f|Y])}{H_{\Phi}(f)}.$$

This implies that given f, $\sup_{\Phi \in \mathcal{F}} \frac{H_{\Phi}(\mathbb{E}[f|Y])}{H_{\Phi}(f)}$ is attained by $\widetilde{\Phi}$. ■

IV. Proof of Theorem 1 for Finite Y

For brevity, given f, we denote

$$\mathbb{H}_f : g \mapsto \frac{H_g(\mathbb{E}[f|Y])}{H_g(f)}.$$

We only need to prove that

$$\sup_{f \in \widetilde{\mathcal{F}}} \sup_{g \in \mathcal{G}} \mathbb{H}_f(g) \leq s^*(X,Y).$$

To prove this, it suffices to prove the following theorem.

Theorem 3. *For binary-finite source pair (X,Y) and given $f \in \widetilde{\mathcal{F}}$, $\sup_{g \in \mathcal{G}} \mathbb{H}_f(g)$ is attained by $g(t) = t$ or $g(t) = 1$.*

Assuming that this theorem holds, we have

$$\begin{aligned}\sup_{f \in \widetilde{\mathcal{F}}} \sup_{g \in \mathcal{G}} \mathbb{H}_f(g) &\leq \sup_{f \in \widetilde{\mathcal{F}}} \max\left\{\mathbb{H}_f(t \mapsto t), \mathbb{H}_f(t \mapsto 1)\right\} \\ &= \max\left\{s^*(X,Y), \rho_{\mathrm{m}}^2(X,Y)\right\} \\ &= s^*(X,Y).\end{aligned}$$

That is, Theorem 1 follows. We next prove Theorem 3.

For a pair of binary r.v.'s (X,Y), denote $u_i := f(x_i), p_i := P_X(x_i), i = 0,1$ and $v_j := \mathbb{E}[f(X)|Y = y_j], q_i := P_Y(y_j), 0 \leq j \leq |\mathcal{Y}| - 1$. WLOG, we assume $u_0 \leq v_0 \leq ... \leq v_{|\mathcal{Y}|-1} \leq u_1$. Then we can write $\mathbb{H}_f(g)$ as

$$\mathbb{H}_f(g) = \frac{\sum_{j=0}^{|\mathcal{Y}|-1} q_j \Phi(v_j)}{p_0 \Phi(u_0) + p_1 \Phi(u_1)}. \tag{15}$$

We first make the following claim.

Claim 1. For $g \in \mathcal{G}$, define a function $\overline{g}(t) = k(t-1) + 1$ for $t \geq 1$ with k satisfying $\overline{\Phi}(u_1) = \Phi(u_1)$. Then we have $\Phi(v) \leq \overline{\Phi}(v)$ for any $v \in [1, u_1]$, where $\Phi(x) = \int_1^x \int_1^s \frac{1}{g(t)} dt ds, \overline{\Phi}(x) = \int_1^x \int_1^s \frac{1}{\overline{g}(t)} dt ds$.

Now we prove this claim. We assume that g is not linear, otherwise, this claim follows trivially. By the concavity of g, we know that one of the following cases holds: 1) $g \geq \overline{g}$ on the whole interval $[1, u_1]$; 2) $g \leq \overline{g}$ on the whole interval $[1, u_1]$; 3) there is a value $b \in (1, u_1)$ such that $g \geq \overline{g}$ on the interval $[1, b]$ and $g \leq \overline{g}$ on the interval $[b, u_1]$. By the assumption $\overline{\Phi}(u_1) = \Phi(u_1)$, we have that only the case 3) could happen. This implies that $h := \frac{1}{\overline{g}} - \frac{1}{g}$ is positive (or zero) on the interval $[1, b]$ and negative (or zero) on the interval $[b, u_1]$. By this, we further have that $s \mapsto \int_1^s h(t)\, dt$ is increasing on the interval $[1, b]$ and decreasing on the interval $[b, u_1]$. Hence one of the following cases holds: 1) $s \mapsto \int_1^s h(t)\, dt$ is positive on the whole interval $[1, u_1]$; 2) there is a value $b' \in (1, u_1)$ such that $s \mapsto \int_1^s h(t)\, dt$ is positive on the interval $[1, b']$ and negative on the interval $[b', u_1]$. By the assumption $\overline{\Phi}(u_1) = \Phi(u_1)$, i.e., $\int_1^{u_1} \int_1^s h(t)\, dt ds = 0$, we have that only the case 2) could happen. Therefore, $x \mapsto \int_1^x \int_1^s h(t)\, dt$ is increasing on the interval $[1, b']$ and decreasing on the interval $[b', u_1]$, and moreover, it is zero at $x = u_1$. It means that $x \mapsto \int_1^x \int_1^s h(t)\, dt$ is positive (or zero) on the whole interval $[1, u_1]$. Hence $\Phi(v) \leq \overline{\Phi}(v)$ for any $v \in [1, u_1]$.

By Claim 1, all $\Phi(v_j)$ with $v_j \geq 1, 0 \leq j \leq |\mathcal{Y}| - 1$ are maximized simultaneously by $\overline{g}$ over all $g \in \mathcal{G}$ under the condition that $\Phi(u_1)$ is fixed.

Similarly, one can show that all $\Phi(v_j)$ with $v_j \leq 1, 0 \leq j \leq |\mathcal{Y}| - 1$ are maximized simultaneously by $\widehat{g}$ over all $g \in \mathcal{G}$ under the condition that $\Phi(u_0)$ is fixed, where $\widehat{g}(t) := k'(t-1) + 1$ for $t \leq 1$ with k' satisfying $\widehat{\Phi}(u_0) = \Phi(u_0)$.

We choose $\ddot{g}(t) = \begin{cases} \overline{g}(t) & t \geq 1 \\ \widehat{g}(t) & t \leq 1 \end{cases}$. Then by (15), $\mathbb{H}_f(\ddot{g}) \geq \mathbb{H}_f(g)$, which implies that the optimal g^* attaining $\sup_{g \in \mathcal{G}} \mathbb{H}_f(g)$ is either linear or two-pieces linear with turning point at $t = 1$. Obviously, $\sup_{f \in \widetilde{\mathcal{F}}} \mathbb{H}_f(g)$ induced by linear g is exactly $s^*(X,Y)$. Hence it suffices to only consider the two-pieces linear case. We next prove that for a two-pieces linear function g with turning point at $t = 1$, the induced $\mathbb{H}(g)$ is no larger than the $\mathbb{H}(\check{g})$ induced by a linear function $\check{g}$. To this end, we need the following lemmas.

For a function $g \in \mathcal{G}$, denote $\overline{g} \in \mathcal{G}$ as a linear function $t \mapsto k(t-1) + 1$ with $0 \leq k \leq 1$ such that $\overline{g} \geq g$. Let $\Phi, \overline{\Phi}$ be respectively the twice-integral of $1/g, 1/\overline{g}$. By definition, any $g \in \mathcal{G}$ is nonnegative and nondecreasing, and g' is nonincreasing.

Hence $\Phi''(t) \geq 0$ and $\Phi'''(t) = (1/g)' = -g'/g^2 \leq 0$. Moreover, $g'(t) \geq k$ for $0 \leq t \leq 1$ and $g'(t) \leq k$ for $t \geq 1$. Obviously, it also holds that $\overline{\Phi}''(t) \geq 0$ and $\overline{\Phi}'''(t) \leq 0$.

Lemma 4. $\frac{\Phi''(t)}{\overline{\Phi}''(t)} = \frac{\overline{g}(t)}{g(t)}$ *is decreasing for* $t \leq 1$ *and increasing for* $t \geq 1$ *and* $\frac{\overline{g}(1)}{g(1)} = 1$.

Proof: Observe that

$$\begin{aligned}\left[\frac{\overline{g}(t)}{g(t)}\right]' &= \frac{\overline{g}'(t)\, g(t) - g'(t)\, \overline{g}(t)}{[g(t)]^2} \\ &= \frac{\overline{g}(t)}{g(t)}\left[\frac{\overline{g}'(t)}{\overline{g}(t)} - \frac{g'(t)}{g(t)}\right]\end{aligned}$$

For $t > 1$,

$$\frac{\overline{g}(t)}{g(t)} = \frac{1 + \int_1^t \overline{g}'(t)\, ds}{1 + \int_1^t g'(t)\, ds} \leq \max\left\{1,\ \sup_{1<s<t} \frac{\overline{g}'(s)}{g'(s)}\right\}$$

Since $\overline{g}' = k$ is constant, and g' is nonincreasing, we have that $\frac{\overline{g}'(s)}{g'(s)}$ is nondecreasing. Hence

$$\sup_{1<s<t} \frac{\overline{g}'(s)}{g'(s)} = \frac{\overline{g}'(t)}{g'(t)} \geq 1,$$

which further implies that for $t > 1$,

$$\frac{\overline{g}(t)}{g(t)} \leq \frac{\overline{g}'(t)}{g'(t)}.$$

Hence $\left[\frac{\overline{g}(t)}{g(t)}\right]' \geq 0$ for $t > 1$.

Similarly, one can prove that $\left[\frac{\overline{g}(t)}{g(t)}\right]' \leq 0$ for $t < 1$. ■

Lemma 5. $\frac{\Phi'(t)}{\overline{\Phi}'(t)}$ *is decreasing for* $t \leq 1$ *and increasing for* $t \geq 1$ *and* $\frac{\Phi'(1)}{\overline{\Phi}'(1)} = 1$.

Proof: Observe that

$$\begin{aligned}\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' &= \frac{\Phi''(t)\, \overline{\Phi}'(t) - \overline{\Phi}''(t)\, \Phi'(t)}{\left[\overline{\Phi}'(t)\right]^2} \\ &= \frac{\overline{\Phi}''(t)}{\overline{\Phi}'(t)}\left[\frac{\Phi''(t)}{\overline{\Phi}''(t)} - \frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]\end{aligned}$$

and $\frac{\Phi''(t)}{\overline{\Phi}''(t)} = \frac{\overline{g}(t)}{g(t)}$.

For $t > 1$,

$$\frac{\Phi'(t)}{\overline{\Phi}'(t)} = \frac{\int_1^t \frac{1}{g(s)} ds}{\int_1^t \frac{1}{\overline{g}(s)} ds} \leq \sup_{1<s<t} \frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}} = \frac{\overline{g}(t)}{g(t)} = \frac{\Phi''(t)}{\overline{\Phi}''(t)}$$

and $\overline{\Phi}'(t), \overline{\Phi}''(t) > 0$. Hence $\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' \geq 0$.

For $t < 1$,

$$\frac{\Phi'(t)}{\overline{\Phi}'(t)} = \frac{\int_t^1 \frac{1}{g(s)} ds}{\int_t^1 \frac{1}{\overline{g}(s)} ds} \leq \sup_{t<s<1} \frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}} = \frac{\overline{g}(t)}{g(t)} = \frac{\Phi''(t)}{\overline{\Phi}''(t)}$$

and $\overline{\Phi}'(t) < 0, \overline{\Phi}''(t) > 0$. Hence $\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' \leq 0$. ■

Lemma 6. $\frac{\Phi(t)}{\overline{\Phi}(t)}$ *is decreasing for* $t \leq 1$ *and increasing for* $t \geq 1$ *and* $\frac{\Phi(1)}{\overline{\Phi}(1)} = 1$.

Proof: Observe that

$$\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]' = \frac{\Phi'(t)\, \overline{\Phi}(t) - \overline{\Phi}'(t)\, \Phi(t)}{\overline{\Phi}^2(t)}$$

and

$$\left[\Phi'(t)\overline{\Phi}(t)-\overline{\Phi}'(t)\Phi(t)\right]'=\Phi''(t)\overline{\Phi}(t)-\overline{\Phi}''(t)\Phi(t)$$
$$=\overline{\Phi}(t)\overline{\Phi}''(t)\left[\frac{\Phi''(t)}{\overline{\Phi}''(t)}-\frac{\Phi(t)}{\overline{\Phi}(t)}\right]$$

For $t>1$,

$$\frac{\Phi''(t)}{\overline{\Phi}''(t)}=\frac{\overline{g}(t)}{g(t)},$$
$$\frac{\Phi(t)}{\overline{\Phi}(t)}=\frac{\int_1^r\int_1^s\frac{1}{g(t)}dtds}{\int_1^r\int_1^s\frac{1}{\overline{g}(t)}dtds}\leq\sup_{1<s<t}\frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}}=\frac{\overline{g}(t)}{g(t)},$$

which implies $\left[\Phi'(t)\overline{\Phi}(t)-\overline{\Phi}'(t)\Phi(t)\right]'\geq 0$. Hence we have

$$\Phi'(t)\overline{\Phi}(t)-\overline{\Phi}'(t)\Phi(t)\geq\Phi'(1)\overline{\Phi}(1)-\overline{\Phi}'(1)\Phi(1)=0$$

That is, $\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]'\geq 0$.

For $t<1$, similarly, we have

$$\left[\Phi'(t)\overline{\Phi}(t)-\overline{\Phi}'(t)\Phi(t)\right]'\geq 0$$

Hence

$$\Phi'(t)\overline{\Phi}(t)-\overline{\Phi}'(t)\Phi(t)\leq\Phi'(1)\overline{\Phi}(1)-\overline{\Phi}'(1)\Phi(1)=0$$

That is, $\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]'\leq 0$. ■

Lemma 7. *For $a_i,b_i,c_i,d_i,i=0,1$ such that $b_i>d_i>0$ and*

$$\frac{a_i}{b_i}\leq\frac{c_i}{d_i},\tag{16}$$

we have

$$\frac{a_0+a_1}{b_0+b_1}\leq\max\left\{\frac{a_0+c_1}{b_0+d_1},\frac{c_0+a_1}{d_0+b_1}\right\}.$$

Proof: Case of $\frac{a_0}{b_0}\geq\frac{a_1}{b_1}$: By (16),

$$\frac{a_1-c_1}{b_1-d_1}\leq\frac{a_1}{b_1}$$

If $\frac{a_0}{b_0}\geq\frac{a_1}{b_1}$, then

$$\frac{a_1-c_1}{b_1-d_1}\leq\frac{a_0}{b_0}$$

That is,

$$b_0(a_1-c_1)\leq a_0(b_1-d_1)\tag{17}$$

By (16) and (17),

$$a_0b_0+b_0(a_1-c_1)+a_1d_1\leq a_0b_0+a_0(b_1-d_1)+c_1b_1$$

That is,

$$\frac{a_0+a_1}{b_0+b_1}\leq\frac{a_0+c_1}{b_0+d_1}$$

Case of $\frac{a_0}{b_0}<\frac{a_1}{b_1}$: Similarly, one can show that if $\frac{a_0}{b_0}<\frac{a_1}{b_1}$, then

$$\frac{a_0+a_1}{b_0+b_1}\leq\frac{c_0+a_1}{d_0+b_1}$$

■

Let $g_1(t):=k_1(t-1)+1$, $g_2(t):=k_2(t-1)+1$, and $g(t):=\begin{cases}g_1(t) & t\leq 1\\ g_2(t) & t>1\end{cases}$, where $k_1\geq k_2$. We next show the following lemma.

Lemma 8. *Let* $U := f(X), V := \mathbb{E}[f(X)|Y]$. *Then*

$$\frac{H_\Phi(V)}{H_\Phi(U)} \leq \max\left\{\frac{H_{\Phi_1}(V)}{H_{\Phi_1}(U)}, \frac{H_{\Phi_2}(V)}{H_{\Phi_2}(U)}\right\}$$

where Φ, Φ_1, Φ_2 *are respectively the twice-integrals of* $1/g, 1/g_1, 1/g_2$.

Proof: For binary sources, denote $u_i := f(x_i), i = 0, 1$ and $v_j := \mathbb{E}[f(X)|Y = y_j], j = 0, 1, ..., N-1$. WLOG, we assume $u_0 \leq v_0 \leq ... \leq v_{N_0-1} \leq 1 \leq v_{N_0} \leq ... \leq v_{N-1} \leq u_1$ for some $1 \leq N_0 \leq N$. By Lemma 6,

$$1 \leq \frac{\Phi(v_j)}{\Phi_2(v_j)} \leq \frac{\Phi(u_0)}{\Phi_2(u_0)} \text{ for } j \leq N_0 - 1,$$
$$1 \leq \frac{\Phi(v_j)}{\Phi_1(v_j)} \leq \frac{\Phi(u_1)}{\Phi_1(u_1)} \text{ for } j \geq N_0.$$

That is,

$$\frac{\Phi(v_j)}{\Phi(u_0)} \leq \frac{\Phi_2(v_j)}{\Phi_2(u_0)} \text{ for } j \leq N_0 - 1,$$
$$\frac{\Phi(v_j)}{\Phi(u_1)} \leq \frac{\Phi_1(v_j)}{\Phi_1(u_1)} \text{ for } j \geq N_0,$$

which further imply

$$\frac{\sum_{j=0}^{N_0-1} P_Y(y_j)\Phi(v_j)}{P_X(x_0)\Phi(u_0)} \leq \frac{\sum_{j=0}^{N_0-1} P_Y(y_j)\Phi_2(v_j)}{P_X(x_0)\Phi_2(u_0)},$$
$$\frac{\sum_{j=N_0}^{N-1} P_Y(y_j)\Phi(v_j)}{P_X(x_1)\Phi(u_1)} \leq \frac{\sum_{j=N_0}^{N-1} P_Y(y_j)\Phi_1(v_j)}{P_X(x_1)\Phi_1(u_1)}.$$

By Lemma 7,

$$\frac{H_\Phi(V)}{H_\Phi(U)} \leq \max\left\{\frac{H_{\Phi_1}(V)}{H_{\Phi_1}(U)}, \frac{H_{\Phi_2}(V)}{H_{\Phi_2}(U)}\right\}.$$

■

Lemma 8 implies that for a given f and a two-pieces linear function g, the induced $\mathbb{H}_f(g)$ is no larger than the one induced by a linear function g. This implies Theorem 1, since the supremum of $\mathbb{H}_f(g)$ over all $f \in \widetilde{\mathcal{F}}$ for a linear function g, is equal to the supremum of $\mathbb{H}_f(g)$ over all $f \in \widetilde{\mathcal{F}}$ for function $g(t) = t$. The latter supremum is exactly $s^*(X, Y)$.

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